

ABL Outreach & Fraud Operations Suite

Agentic AI System — Cold Outreach, Fraud Flagging & Automated Charity Consent Workflow

Document Type	Software Requirements & Design Specification
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Modules Covered in This Document

- 1 Cold Email Outreach Agent
- 2 Transaction Fraud Flagging Agent
- 3 Automated Charity Consent Workflow Agent (late-fee-to-donation consent handling)

This specification is written to be implementation-agnostic: it defines what each module must do, how modules interact, and what data and behaviour are required, without prescribing a specific programming language, framework, or hosting choice. Technology selection is a separate decision left to the implementing engineer.

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1. Introduction

1.1 Purpose

This document specifies an agentic AI system comprising three coordinated modules used in outbound communication and fraud-risk operations for a retail banking context. The system is designed to demonstrate autonomous, multi-step reasoning and tool-use over synthetic banking data, modelled after real ABL operational practices observed during the ABIP internship (Aug-Sep 2026), including the bank's existing late-payment-to-charity consent process. The intent is to show, end to end, that an agent can be trusted to carry a multi-turn operational workflow to a correct, auditable conclusion without a human in the loop for every step.

1.2 Stakeholders

Role	Interest
Sir Affan Wahid	Internship supervisor and primary reviewer; evaluates technical soundness and fidelity to real ABL process.
Daniyal Saqib	Sole developer; responsible for design, implementation, and demonstration.
Hypothetical Charity/CSR Dept.	Receiving department for consented charity-fee cases in the real-world process this module mirrors.
Hypothetical Collections Dept.	Receiving department for declined or escalated cases.

1.3 Scope

In scope for this delivery:

- Autonomous drafting and personalization of outbound emails against a mock prospect/customer list
- Autonomous detection and flagging of anomalous transactions from a synthetic transaction feed
- A fully agentic, multi-turn consent workflow modelled on ABL's real late-fee-to-donation process, including follow-up and escalation logic
- A single orchestration layer coordinating all three modules with a shared logging and audit trail

Explicitly out of scope:

- Any integration with ABL's live email, core banking, or CRM systems
- Real customer data of any kind
- Actual delivery of emails to real people outside of a demo/sandbox environment
- Legal or regulatory sign-off — this is a prototype demonstrating architecture and reasoning, not a production-ready compliance system

1.4 Data Constraint

Allied Bank Limited does not, and will not, provide real customer, transaction, or account data for this project. All data used for development, testing, and demonstration is synthetically generated and stored locally. No production ABL system is connected to, queried, or modified at any point. Any resemblance to real account numbers, balances, or customer names in sample data is coincidental and clearly marked as mock data throughout the build and demo. This constraint applies uniformly across every module described in this document and is treated as a hard requirement, not a preference.

2. System Architecture

The system follows a layered agentic architecture. A single orchestrator receives a task, determines which sub-agent should handle it, and passes structured context to that agent. Each agent can call defined tools (data lookups, template rendering, mock message sending) and must return a structured result the orchestrator logs before taking the next action. All state transitions are persisted so any case can be audited end-to-end.

Layer	Responsibility
Presentation Layer	Operator-facing view showing campaign status, flagged transactions, and the live status of every open consent case.
Orchestration Layer	Routes each incoming task to the correct sub-agent, maintains conversation/case state, and enforces the retry and escalation rules defined in Section 4.
Cold Outreach Agent	Generates and personalizes outbound email drafts per prospect.
Fraud Flagging Agent	Scores each transaction and explains flags in plain language.
Charity Consent Agent	Drives the multi-turn consent state machine described in Section 4.
Service Layer	Exposes the actions defined in Section 5 to the presentation layer and to the agents themselves.
Data Layer	Persists prospects, transactions, consent cases, and the full audit log.

2.1 High-Level Data Flow

- 1 A trigger event occurs (new prospect added / new transaction ingested / loan enters late-fee status).
- 2 The Orchestration Layer creates a task record and routes it to the relevant agent.
- 3 The agent reasons over the task using available tools and produces a draft output (email, fraud flag + explanation, or consent-case next action).
- 4 Output is written to storage and surfaced on the dashboard for operator review.
- 5 For the Charity Consent Agent specifically, the case remains open and is re-evaluated on a schedule until it reaches a terminal state (see Section 4).

2.2 Design Principles

- **Separation of concerns** — each agent owns exactly one responsibility and never reaches into another agent's data directly.
- **Everything is logged** — no agent action is allowed to happen silently; every decision produces an entry in the audit trail.
- **Fail closed, not open** — when an agent is uncertain, the system defaults to the safest, most conservative outcome rather than guessing.

3. Data Model (Mock / Synthetic)

All entities below are populated with generated data at seed time. No entity is ever sourced from a live ABL system. Field names describe intent, not a fixed schema — the implementing engineer may adapt types and storage as appropriate for their chosen approach.

Prospect

Represents a mock outreach target used by the Cold Outreach Agent.

Field	Notes
prospect_id	Primary key, uniquely identifies the record
name	Mock generated
company	Mock generated
email	Sandbox address only — never a real inbox
outreach_status	One of: not_started / drafted / sent / responded

Transaction

Represents a single mock banking transaction scored by the Fraud Flagging Agent.

Field	Notes
transaction_id	Primary key
account_id	Mock account reference, never tied to a real ABL account
amount	Decimal value of the transaction
timestamp	Date and time the transaction occurred
fraud_score	Agent-assigned value from 0.0 (no concern) to 1.0 (high concern)
flag_reason	Agent-generated plain-language explanation for the score

CharityConsentCase

Represents one loan's journey through the consent workflow described in Section 4.

Field	Notes
case_id	Primary key
loan_id	Mock loan reference
state	Current position in the state machine — see Section 4.3
follow_up_count	Integer from 0 to 3, tracking how many follow-ups have been sent
last_contacted_at	Date and time of the most recent message sent to the customer
resolution	Final outcome once terminal: consented / declined / escalated_no_response / escalated_ambiguous

4. Module Deep-Dive: Automated Charity Consent Workflow

This module is modelled directly on Allied Bank's real-world practice: when a loan customer becomes late on a payment, they are sent a consent request asking whether the resulting late-payment fee should be directed to charitable donation. This is the single most important module in this application from a demonstration standpoint, because it is the one most directly traceable to an actual, currently-running ABL process. The agent must handle four distinct response categories and apply a bounded follow-up policy before escalating unresolved cases to a human team.

4.1 Response Categories

Response	Immediate Action
Yes (explicit consent)	Case closes as 'consented'; routed to the Charity/CSR department for processing.
No (explicit decline)	Case closes as 'declined'; routed to the Collections/Fees department for standard fee handling.
Ambiguous / unclear reply	Agent drafts and sends a clarification message; does not close the case.
No response received	Agent drafts and sends a follow-up message after a defined wait period; does not close the case.

4.2 Follow-Up and Escalation Policy

- Up to 3 follow-up/clarification messages are sent per case, regardless of whether the trigger was silence or ambiguity.
- Each follow-up increments the follow-up count; a fresh wait window starts after each send.
- If a clear Yes or No response arrives at any point, the case immediately closes on that path — no further follow-ups are sent.
- If the follow-up count reaches 3 with no clear resolution, the case is force-closed with resolution 'escalated_no_response' or 'escalated_ambiguous' and routed to the relevant department for manual handling.
- The policy is enforced as fixed program logic, not left to the agent's discretion — the agent may decide what to say in a follow-up, but never whether or how many follow-ups are sent.

4.3 State Machine Summary

State	Entry Condition	Exit Condition
Awaiting Response	Initial consent message sent	Response received, or wait window expires
Follow-Up 1 / 2 / 3	Ambiguous reply or timeout in prior state	Clear response received, or window expires again
Resolved: Consented	Explicit Yes received in any state	Terminal — routed to Charity/CSR
Resolved: Declined	Explicit No received in any state	Terminal — routed to Collections
Escalated: Unresolved	3rd follow-up expires without clear response	Terminal — routed to Manual Review

4.4 Worked Example

A loan enters late-fee status on Day 0. The initial consent message is sent and the case enters 'Awaiting Response'. On Day 3 the customer replies 'not sure, can you explain more?' — this is classified as ambiguous, so a clarification message is sent and the case moves to 'Follow-Up 1'. On Day 6, no reply has arrived; the wait window expires, so a second follow-up is sent and the case moves to 'Follow-Up 2'. On Day 9 the customer replies 'yes, that's fine' — a clear Yes — so the case closes immediately as 'Resolved: Consented' and is routed to the Charity/CSR department, even though only 2 of the allowed 3 follow-ups were used.

5. Interface Design

The interfaces below describe the actions each module must expose, independent of how they are ultimately implemented (REST, RPC, direct function calls, etc.).

Action	Description
Draft outreach email	Generate a personalized outreach draft for a given prospect.
List flagged transactions	Return all transactions currently flagged by the fraud agent.
Rescan transaction	Re-run fraud scoring on a specific transaction.
Open consent case	Create a new consent case for a loan entering late-fee status.
Record case response	Record an incoming response and advance the state machine.
Fetch case history	Return the full audit trail of a consent case.

6. Non-Functional Requirements

Category	Requirement
Auditability	Every state transition, agent decision, and message sent must be logged with a timestamp and a short reasoning summary.
Determinism	The follow-up/escalation policy (max 3 follow-ups) must be enforced by fixed logic, not left to model discretion.
Data isolation	No component may connect to a live ABL system or hold real customer PII.
Explainability	Every fraud flag must include a human-readable reason, not just a numeric score.
Recoverability	If the system restarts mid-case, in-progress consent cases must resume from their last recorded state, not reset.

7. Risks & Mitigations

Risk	Likelihood	Mitigation
Mock message delivery mistaken for real outreach	Low	Use a fully sandboxed messaging channel exclusively; never a live relay to real inboxes.
Ambiguous-response classification errors	Medium	Keep classification conservative — default unclear cases to 'ambiguous' rather than guessing Yes/No.
Scope creep across three sub-agents	Medium	Build the Charity Consent Agent first (highest demo value); treat the other two as time-permitting extensions.
Follow-up timing logic drifts from the stated policy during development	Low	Cover the state machine with automated tests for every transition in Section 4.3 before building the UI.

8. Success Criteria

The following criteria define what a working demonstration must show by the end of the project window. These are the minimum bar for the build to be considered complete, independent of any stretch goals.

#	Criterion
1	A demo can walk a single consent case through all five terminal/non-terminal states live, on request.
2	The follow-up count never exceeds 3 for any case, verified against the audit log.
3	Every fraud flag shown in the demo includes a plain-language reason, not only a score.
4	The full audit trail for any case can be retrieved and read end-to-end without gaps.

9. Project Timeline (27 Aug - 25 Sep 2026)

Week	Dates	Milestone
Week 1	27 Aug - 02 Sep	Data model, mock data generator, orchestrator skeleton
Week 2	03 Sep - 09 Sep	Charity Consent Agent: state machine + follow-up scheduler complete
Week 3	10 Sep - 16 Sep	Fraud Flagging Agent + Cold Outreach Agent built
Week 4	17 Sep - 23 Sep	Dashboard integration, end-to-end testing, audit log review
Buffer	24 Sep - 25 Sep	Polish, demo rehearsal, documentation finalization

10. Assumptions

- Sir Affan's review is a milestone check-in, not a formal client acceptance process — scope may be adjusted after each week based on feedback.
- A sandboxed messaging channel is available for the duration of the project so real inboxes are never used.
- The mock data generator can produce enough volume and variety to make fraud scoring and follow-up timing meaningfully demonstrable, not just trivially correct on a handful of hand-picked examples.
- No department integration is required for the demo — 'routed to Collections' etc. means the case is labelled and logged as such, not that a real handoff occurs.

11. Glossary of Terms

Term	Definition
Agent	A component that autonomously reasons over a task, may call tools, and produces a structured decision or output.
Orchestrator	The component that decides which agent handles an incoming task and tracks its progress.
Terminal state	A state in the consent workflow from which no further transition occurs — the case is closed.
Mock / synthetic data	Data generated for development and demo purposes that does not represent any real person or account.
Audit trail	The complete, timestamped record of every decision and action taken on a given case.